INFORMATIK

RPTU Rheinland-Pfälzische Technische Universität Kaiserslautern Landau

# Model-Based Development of a System Architecture for Vehicle Platforms

Master Thesis

by

Stawan Chandrashekhar Kulkarni

Chair of Design of Cyber-Physical Systems

Department of Computer Science

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# Agenda

- Introduction
- SysMD
- Methodology
- Implementation
- Conclusion

# Agenda

- **Introduction**
  - Motivation
  - Related Work
  - SysML v2
- SysMD
- Methodology
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# Motivation

- “Today, the **software is decentralized in a lot of black boxes**, and we buy it from suppliers based on specifications, but **when we want a change** and you have to talk to suppliers, **it is too slow**,”

  - Hakan Samuelsson, CEO Volvo Group [1]

- GENIAL! And KI4Boardnet
  - Early analysis of microelectronic components[2]
  - Shared stakeholders understanding [2]
  - Performance and Procedures for: E/E Architecture, Components, Wiring Harness, Control units [3]

# Motivation

- The problem of Automotive Systems Engineering practices
  - Managing complexity of product [4]
  - Involvement of stakeholders in value chains [4]
  - Effective change management [4]
- Specification phase
  - Requirements, Use-cases
  - Errors lead to project failure [5]
  - Relative cost of bug fixing is very low [5]
  - Completeness and Consistency [6]

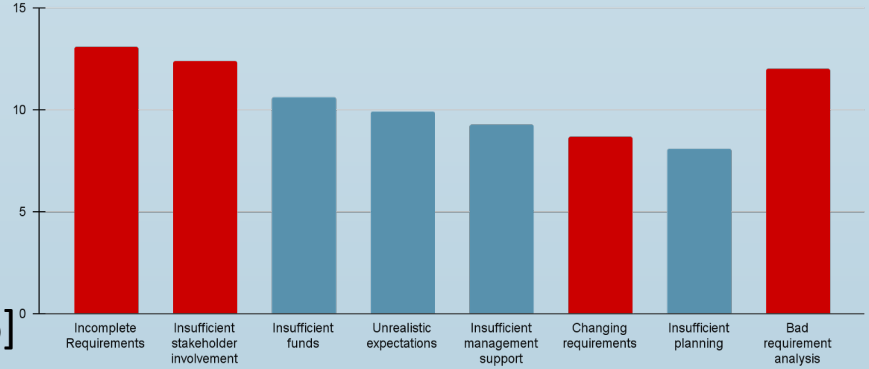


Figure 1: Reasons for "Failed" or "Challenged" Projects [5]

# Motivation

- Automotive Boardnet
  - Sensors
  - Control Units
  - Bus Systems
  - Power Systems
- Thesis objective
  - Development of logical and technical architecture specifications
  - Using resources from the feature list from the project KI4Boardnet
  - Contributing to OpenCar model suite:
    - Analysis of Features, Topologies, Sensors, Computing Hardwares etc
    - Network architecture considering geometric aspects

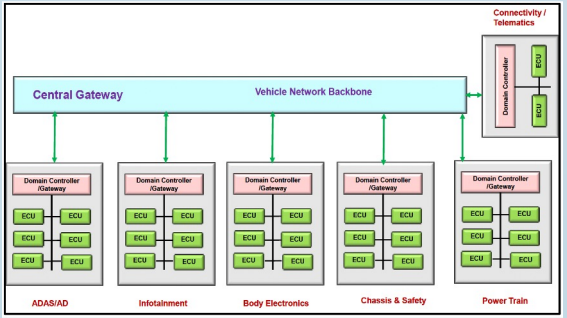


Figure 2: Vehicle Network Architecture [7]

# Related Work

- Methodologies [8]
  - IBM Harmony for SE
  - Pattern based SE
  - Weilkins' SysMOD+
  - Object Process Modeling
  - OOSEM
- Modeling Languages
  - UML [9]
  - OPML [8]
  - AADL [8]
- Tools[10]
  - Cameo Systems Modeler
  - ARTiSAN studio
  - Gaphor
  - Enterprise Architect

# SysML v2

- What is SysML?
  - Requirements,Specification, Use cases, and Verification and Validation [7]
  - Systems involving Hardware, Software, Information, Processes, People [4]
- SysML v2 [11]
  - KerML: New metamodel
  - Robust visualisations
  - Textual modeling language
  - API: access, query, update, navigate

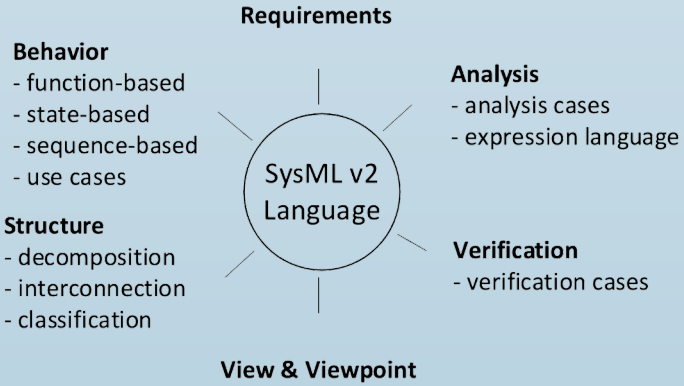


Figure 3: SysML v2 Capabilities [11]

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# SysMD

- SysMD Notebook [12]
  - Static Analysis of Specifications
  - UI, Compiler, Analysis (Constraint propagation)
  - Connected to AGILA backend using REST API
  - Documentation: Tables, Figures, link, Markdown Text
  - Model: Definition, Instances, Values, Constraints, Relationships

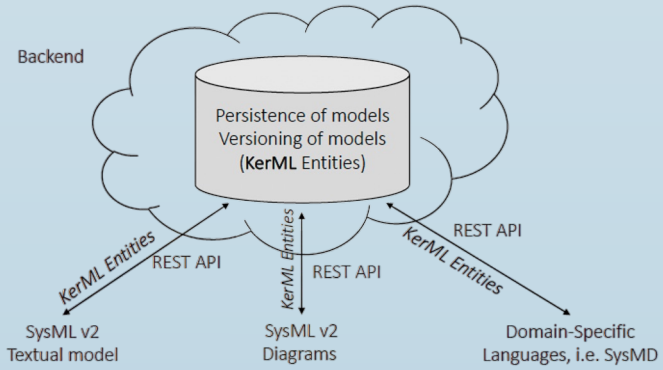


Figure 4: Integration into SysML ecosystem [12]

# SysMD

- Lexical Insights [11]

| | |
|---|---|
| **Literals** | String, Real, Integer, Boolean |
| **Keywords** | One, all, and, or, hasA, isA, defines |
| **Math Operators** | +,-,*,/,^, |
| **Math Functions** | sqrt(x),sqr(x), ln(x),pow(x,y), exp(x), max(x,y), min(x,y) |
| **Aggregation functions** | sumOverParts, sumOverSubclasses, productOverParts, productOverSubclasses |

- SysMD Modeling language
  - Subset of SysML v2 with extensions [11]
  - Domain specific for boardnet [12]
  - Statements can be given in any order [11]
  - Incremental compilation [11]
  - Ranges: uncertainties as lower and upper limits [12]
  - Vectors instead of normal values, dot, cross, angle operations [12]

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# Approach

- Object Oriented Systems Engineering Methodology (OOSEM) [13]
  - SysMD as modeling tool
  - Sticks to ISO/IEC/IEEE15288:2015
  - Leverages SysML v2 concepts
  - Tool, Language independent
  - Functional decomposition
- Feature list
  - Start from feature description
  - Identify functions, components
  - Component pin list

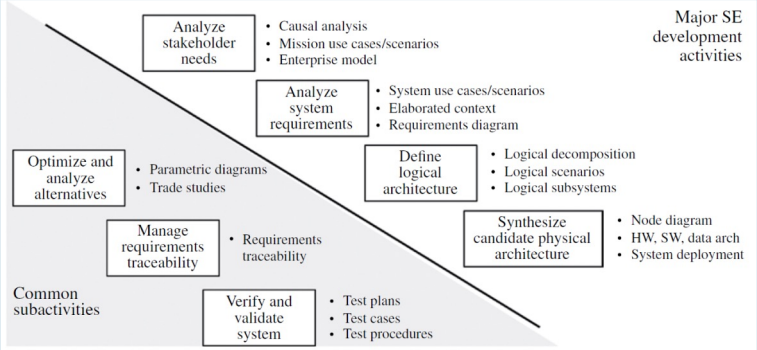


Figure 5: Activities and Artifacts in OOSEM [10]

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  - Future Scope

# Conclusion

- Specification models for architectures of an Adaptive Cruise Control system were developed
- Focus was mainly on: Components, Functions in an E/E system along with Mappings, Constraints
- Packages for AUTOSAR, Requirements were developed
- The model act as template for modeling other systems in the vehicle
- A geometric perspective of Installation spaces was developed
  - Determination of weight distribution within vehicle
- Signifies capabilities of SysML v2 models and SysMD in SE processes

# Future Scope

- Software Components
  - Algorithms: PID Control, Perceptron, BiQuad filter
  - Models for signals
  - Control signal parameters
  - Interval arithmetic [15] , Ranges would be crucial
  - Detection of possible overflow, underflow
- Geometric Models
  - Model for points, lines,surfaces,planes
  - Constraints for parallelism, perpendiculars, tangent
  - 2D elements: Circle, Triangles, Squares, and So on..
  - Possibility for CAD vendors to integrate SysML v2 models

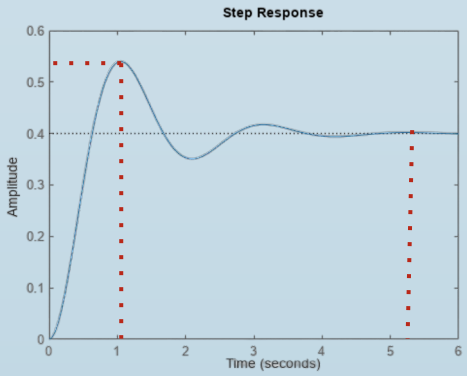


Figure 6: Step response of PID [14]

# Thank you!

# References


[1] Trudell,Craig: Volvo's New CEO says carmakers must smarten up on software. 2022. – https://europe.autonews.com/automakers/volvos-new-ceo-says-automaker-must-be-smarter-software [Online;accessed 17-January-2024]

[2] Für Bildung und Forschung, Bundesministerium: Gemeinsame Elektronik-Roadmap für Innovationen der automobilen Wertschöpfungskette (GENIAL!). 2018. – https://www.elektronikforschung.de/projekte/genial-1 [Online; accessed 17-January-2024]

[3] KI4Boardnet: E/E Entwicklung für Energie und Datenboardnetze. 2023 - https://www.edacentrum.de/ki4boardnet/ [Online; accessed 17-January-2024]

[4] Freidenthal,Sanford and Seidewitz, Ed: SysML v2: Where we are and how we got here. 2023. – https://www.omg.org/pdf/SysML-v2-Overview.pdf [Online; accessed 17-January-2024]

[5] Dörr, Jörg: Lecture notes in Requirements Engineering. University of Kaiserslautern-Landau. November 2022

[6] Grimm, Christoph. Lecture notes in Automotive Software and Systems Engineering. June 2021

# References


[7] Mody, Mihir ; Jones, Jason ; Chitnis, Kedar ; Sagar, Rajat ; Shurtz, Gregory ; Dutt, Yashwant ; Koul, Manoj ; Biju, MG ;Dubey, Aish: Understanding vehicle e/e architecture topologies for automated driving: System partitioning and tradeoff parameters. In: Electronic Imaging 2018 (2018), Nr. 17, S. 358–1

[8] Alai, Shashank P.: Evaluating Arcadia/Capella Vs OOSEM/SysML for System Architecture Development, Purdue University, masterthesis, 2019

[9] Object Management Group: INTRODUCTION TO OMG'S UNIFIED MODELING LANGUAGE. 2005. – http://www.uml.org/what-is-uml.html [Online; accessed 28-May-2023]

[10] Weilkins, Tim: SysML v2 modeling tools. 2022. – https://mbse4u.com/2022/03/09/sysml-v2-modeling-tools/ [Online; accessed 08-January-2024]

[11] Grimm,Christoph and Ratzke, Axle and Post, Sebastian and Heermann, Hagen and Koch, Johannes : SysML v2 An overview with SysMD Demonstration- Design and VerificationConference and Exhibition, November 14-15, 2023, Munich, Germany.

[12] Post, Sebastian ; Grimm, Christoph: Co-Design of Automotive Boardnet Topology and Architecture. In: Design and Verification Conference and Exhibition, November 14-15, 2023, Munich, Germany VDE, 2023, S. 42–59

# References


[13] Object Management Group: INCOSE Object Oriented Systems Engineering Method. 20011. – https://omgwiki.org/MBSE/doku.php?id=mbse:incoseoosem [Online; accessed 19-January-2024]

[14] Matlab: Step response of a dynamic system 2023. – https://de.mathworks.com/help/control/ref/dynamicsystem.step.html [Online; accessed 19-January-2024]

[15] Kulisch, Ulrich. "Mathematics and speed for interval arithmetic: A complement to ieee 1788." ACM Transactions on Mathematical Software (TOMS) 45.1 (2019): 1-22.